DefenderPi - AI Assisted Inline Network Defense System for Small and Edge Networks Using Raspberry Pi

Executive Summary

DefenderPi is a Raspberry Pi-based inline cyber defense platform that combines intrusion prevention, DNS security, behavioral anomaly detection, automated containment, and SOC-style monitoring. The system operates as a security gateway, inspecting network traffic in real time and automatically responding to malicious activity while achieving real-time detection and automated containment on edge hardware.

Project Overview

Problem

Most home networks, small businesses, and IoT environments lack visibility into network activity and rely on consumer-grade routers that provide minimal protection against modern cyber threats.
Enterprise security solutions such as firewalls, IDS/IPS platforms, SIEM systems, and threat intelligence platforms are often expensive, complex, and resource-intensive.
As a result, small networks remain vulnerable to:

- Port scanning
- ARP spoofing attacks
- Rogue DHCP servers
- DNS-based attacks
- Brute-force attempts
- Network reconnaissance
- Malicious outbound communication
- IoT compromise
  DefenderPi was developed to address this gap by providing a practical, affordable, and deployable cyber defense solution capable of detecting and containing threats in real time.

Threat Model

DefenderPi was designed to address common threats encountered in home networks, small offices, educational laboratories, and IoT environments.

Threat	Impact	Detection Method
Port Scanning	Network Reconnaissance	Suricata IPS
Brute Force Attempts	Credential Compromise	Suricata IPS
ARP Spoofing	Man-in-the-Middle Attacks	arpwatch
Rogue DHCP	Traffic Redirection	DHCP Monitor
DNS Abuse	Malicious Domain Access	Pi-hole + Unbound
Suspicious Device Behaviour	Unknown Threat Activity	Isolation Forest + KMeans
Malicious Outbound Connections	Data Exfiltration	Multi-Layer Correlation

The platform combines signature-based detection, behavioral analytics, and network integrity monitoring to provide layered protection against both known and unknown threats.

System Architecture

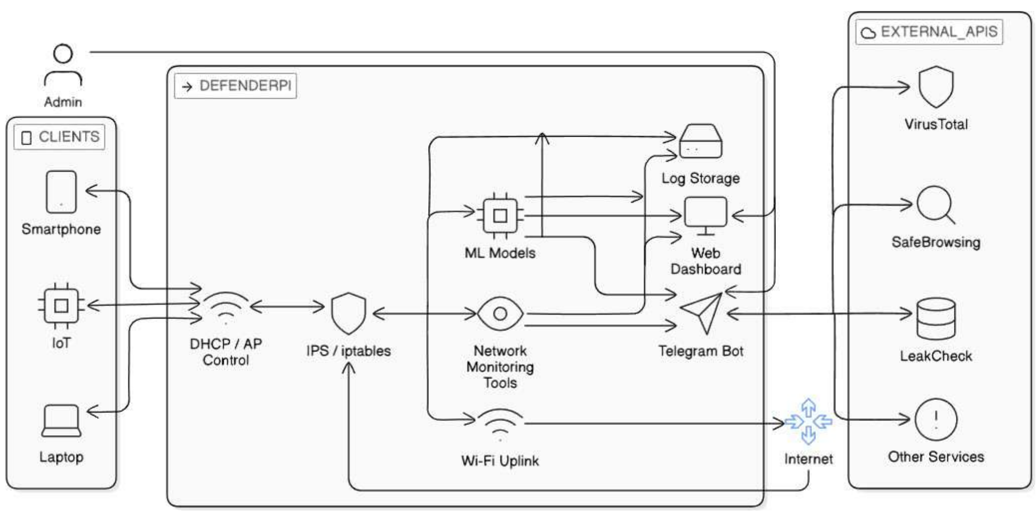


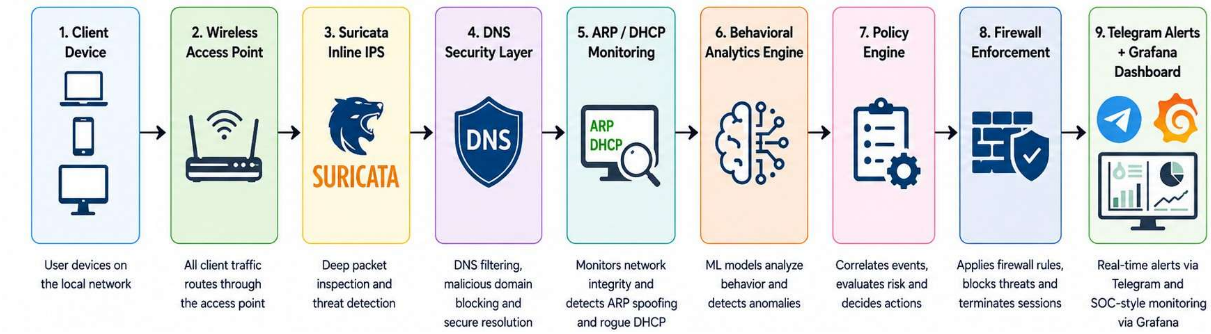
Figure 1. DefenderPi Multi-Layer Security Architecture

Description

The DefenderPi architecture places the Raspberry Pi between client devices and the internet, ensuring that all traffic passes through multiple security controls including intrusion prevention, anomaly detection, DNS security, firewall enforcement, monitoring services, and external threat intelligence integrations.

Detection & Response Pipeline

DefenderPi follows a multi-stage detection and response workflow that correlates events from multiple security layers before initiating enforcement actions.



This architecture ensures that enforcement decisions are based on multiple independent security signals rather than isolated alerts, reducing false positives while enabling rapid threat containment.

Technology Stack

Layer	Technologies
Infrastructure	Raspberry Pi 4, Raspberry Pi OS Lite
Network Security	Suricata, Pi-hole, Unbound, arpwatch
Traffic Control	hostapd, dnsmasq, iptables, ipset
Machine Learning	Isolation Forest, K-Means
Monitoring&Alerting	Grafana, Redis, Telegram Bot
Threat Intelligence	VirusTotal, AbuseIPDB, Google Safe Browsing, URLhaus, HaveIBeenPwned
Development	Python, Bash, systemd

Hardware Platform

Component	Cost (₹)
Raspberry Pi 4 (4GB)	5,232
Wi-Fi Adapter	1,109
Storage & Accessories	1,370
Total	**7,711**

DefenderPi delivers intrusion prevention, DNS security, behavioral analytics, and automated threat response on hardware costing less than **₹8,000 (~$90 USD)**.

Core Security Capabilities

Inline Intrusion Prevention System

Suricata operates in NFQUEUE inline mode and performs deep packet inspection before traffic is forwarded.
The system uses small LAN optimized detection rules to identify:

- Port scans
- Reconnaissance activity
- Brute-force attacks
- Protocol anomalies
- Exploit attempts
- Known attack signatures

  Unlike traditional IDS deployments, DefenderPi can automatically block malicious traffic in real time.

Network Integrity Monitoring

ARP Spoofing Detection

DefenderPi continuously monitors IP-to-MAC mappings using arpwatch.
When suspicious changes occur, the platform identifies potential Man-in-the-Middle attacks and generates immediate security alerts.

```
12:35:06 From: arpwatch (Arpwatch defenderpi)
12:35:06 To: root@localhost
12:35:06 Subject: flip flop (192.168.50.76) wlan1
12:35:06
12:35:06             hostname: <unknown>
12:35:06           ip address: 192.168.50.76
12:35:06            interface: wlan1
12:35:06     ethernet address: 10:f6:0a:30:0e:71
12:35:06      ethernet vendor: Intel Corporate
12:35:06 old ethernet address: 56:a6:b9:e2:8a:55
12:35:06  old ethernet vendor: <unknown>
12:35:06            timestamp: Sunday, October 19, 2025 12:35:06 +0530
12:35:06   previous timestamp: Sunday, October 19, 2025 12:35:03 +0530
12:35:06                delta: 3 seconds
12:35:09
```

```
propi@defenderpi:~ $ arp -n
Address                  HWtype  HWaddress           Flags Mask            Iface
192.168.50.59            ether   10:f6:0a:30:0e:71   C                     wlan1
192.168.1.72             ether   10:f6:0a:30:0e:71   C                     wlan0
192.168.50.83                    (incomplete)                              wlan1
192.168.0.1              ether   10:f6:0a:30:0e:71   C                     wlan1
192.168.1.1              ether   64:fb:92:65:e4:44   C                     wlan0
192.168.50.76            ether   56:a6:b9:e2:8a:55   C                     wlan1
172.17.0.2               ether   02:42:ac:11:00:02   C                     docker0
propi@defenderpi:~ $ arp -n
Address                  HWtype  HWaddress           Flags Mask            Iface
192.168.50.59            ether   10:f6:0a:30:0e:71   C                     wlan1
192.168.1.72             ether   10:f6:0a:30:0e:71   C                     wlan0
192.168.50.83                    (incomplete)                              wlan1
192.168.0.1              ether   10:f6:0a:30:0e:71   C                     wlan1
192.168.1.1              ether   64:fb:92:65:e4:44   C                     wlan0
192.168.50.76            ether   10:f6:0a:30:0e:71   C                     wlan1
172.17.0.2               ether   02:42:ac:11:00:02   C                     docker0
```

Figure 3. ARP Spoofing Detection and Alert Generation

Rogue DHCP Detection

Unauthorized DHCP servers can redirect network traffic and compromise connected devices.

DefenderPi monitors DHCP activity using packet inspection techniques and identifies rogue DHCP responses before they impact users

```
propi@defenderpi:/var/log/defenderpi $ sudo /usr/local/bin/defenderpi-test-rogue.py wlan1
Sending rogue OFFER on wlan1 -> server: 192.168.250.99
Sending rogue ACK on wlan1
Done.
propi@defenderpi:/var/log/defenderpi $ sudo tail -n 200 dhcp-wlan1.log
  TIME: 2025-10-19 11:46:50.697
    IP: 192.168.250.99 (2:0:0:0:0:99) > 255.255.255.255 (ff:ff:ff:ff:ff:ff)
    OP: 2 (BOOTPREPLY)
 HTYPE: 1 (Ethernet)
  HLEN: 6
  HOPS: 0
   XID: 00000000
  SECS: 0
 FLAGS: 0
CIADDR: 0.0.0.0
YIADDR: 192.168.250.100
SIADDR: 192.168.250.99
GIADDR: 0.0.0.0
CHADDR: 35:36:61:36:62:39:65:32:38:61:35:35:00:00:00:00
 SNAME: .
 FNAME: .
OPTION:  53 (  1) DHCP message type         2 (DHCPOFFER)
OPTION:  54 (  4) Server identifier         192.168.250.99
---------------------------------------------------------------------------

  TIME: 2025-10-19 11:46:51.841
    IP: 192.168.250.99 (2:0:0:0:0:99) > 255.255.255.255 (ff:ff:ff:ff:ff:ff)
    OP: 2 (BOOTPREPLY)
 HTYPE: 1 (Ethernet)
  HLEN: 6
  HOPS: 0
   XID: 00000000
  SECS: 0
 FLAGS: 0
CIADDR: 0.0.0.0
YIADDR: 192.168.250.100
SIADDR: 192.168.250.99
GIADDR: 0.0.0.0
CHADDR: 35:36:61:36:62:39:65:32:38:61:35:35:00:00:00:00
 SNAME: .
 FNAME: .
```

Figure 4. Rogue DHCP Detection Module

DNS Security Layer

DNS protection is implemented using Pi-hole and Unbound.
The platform blocks:

- Malicious domains
- Phishing infrastructure
- Advertisement networks
- Tracking services
- Known malicious hosts
  All DNS traffic is transparently redirected through the filtering engine, enabling complete visibility into DNS activity while maintaining secure recursive resolution and DNSSEC validation.

```
propi@defenderpi: /var/log    Command Prompt
Microsoft Windows [Version 10.0.26100.6725]
(c) Microsoft Corporation. All rights reserved.

C:\Users\edwin>nslookup ads.doubleclick.net
DNS request timed out.
    timeout was 2 seconds.
Server:  UnKnown
Address:  1.1.1.1

*** UnKnown can't find ads.doubleclick.net: Non-existent domain

C:\Users\edwin>nslookup malicious.example.com
Server:  one.one.one.one
Address:  1.1.1.1

*** one.one.one.one can't find malicious.example.com: Non-existent domain

C:\Users\edwin>
```

Figure 5. DNS Filtering Validation

Behavioral Analytics & Anomaly Detection

DefenderPi incorporates a lightweight behavioral analytics engine designed to identify suspicious activity that may not trigger traditional signature-based detection systems. Behavioral records are generated from per-device network activity using rolling observation windows and engineered traffic features.

K-Means Behavioral Profiling

K-Means clustering is used to establish baseline communication patterns for connected devices.
The model analyzes:

- Traffic Volume, Destination Diversity, Protocol Usage, Traffic Stability, Communication Density
  Engineered Features:
- flows_per_sec, bytes_per_flow, bytes_per_sec, traffic_balance. dst_ip_density, dst_port_density tcp_ratio, udp_ratio, flow_stability
  Principal Component Analysis (PCA) is applied before clustering to reduce feature correlation and improve cluster quality.

Isolation Forest Anomaly Detection

Isolation Forest identifies short-term behavioral anomalies and deviations from normal device activity.
The model evaluates:

- Flow Volume, DNS Activity, Destination Diversity, Traffic Burst Behaviour, Temporal Deviations
  Engineered Features:
- flows_count, bytes_total, unique_dst_ips, unique_dst_ports, flow_burst_ratio, flow_delta, dns_delta, dst_spread_ratio, port_spread_ratio, traffic_skew, dns_queries, dns_entropy_avg, dns_anomaly_ratio

Model Evaluation

Model	Precision	Recall	F1 Score
Isolation Forest	93.75%	78.95%	85.71%
KMeans	95.00%	100.00%	97.44%
Fusion Engine	95.00%	100.00%	97.44%

False Positive Rate: 5.26%
The combined behavioral analytics framework achieved complete detection of evaluated scanning, brute-force, and denial-of-service attack scenarios while maintaining a low false positive rate.

Policy Engine and Automated Response

DefenderPi correlates events from:

- Suricata
- DNS Security Layer
- ARP Monitoring
- DHCP Monitoring
- Behavioral Analysis Models
  Instead of acting on individual alerts, a centralized policy engine evaluates confidence scores and contextual information before initiating enforcement.
  Possible actions include:
- Alert Generation
- Rate Limiting
- Temporary Quarantine
- Permanent Blocking
- Session Termination
  This approach reduces false positives while maintaining rapid threat containment.

Reputation-Based Escalation Engine

DefenderPi includes a reputation-based enforcement engine that tracks repeated malicious activity across Suricata, network monitoring, and behavioral analytics modules. Instead of permanently blocking hosts after a single event, the system applies progressive containment based on offender history, reducing false positives while improving long-term threat containment.

Offense Count	Enforcement Action
1–2	5-minute quarantine
3	15-minute quarantine
4	1-hour quarantine
5	6-hour quarantine
6+	Permanent block

Security events are correlated across multiple detection sources, allowing DefenderPi to automatically escalate enforcement actions against recurring threats while maintaining normal network availability for legitimate users

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Automated Firewall Enforcement

Dynamic containment is implemented using:

- iptables
- ipset
- conntrack

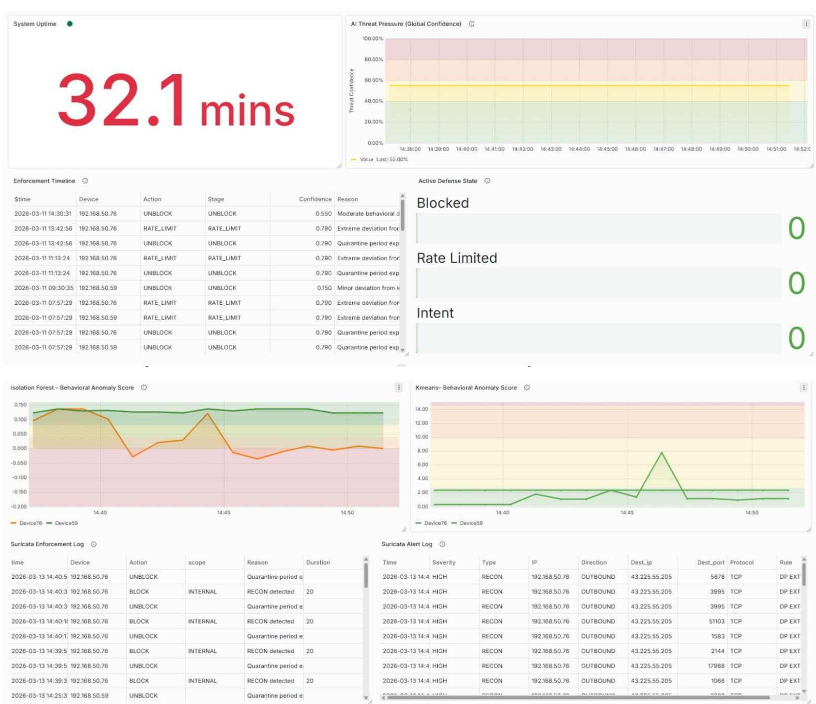
Malicious hosts are automatically quarantined and active attack sessions are terminated immediately.

Security Operations Dashboard

DefenderPi provides centralized monitoring through Grafana dashboards.
Administrators can view:

- Connected devices
- Security alerts
- Network activity
- System health
- Threat detections
- Resource utilization
- Enforcement actions

  The dashboard provides a SOC-style monitoring experience for real-time situational awareness.



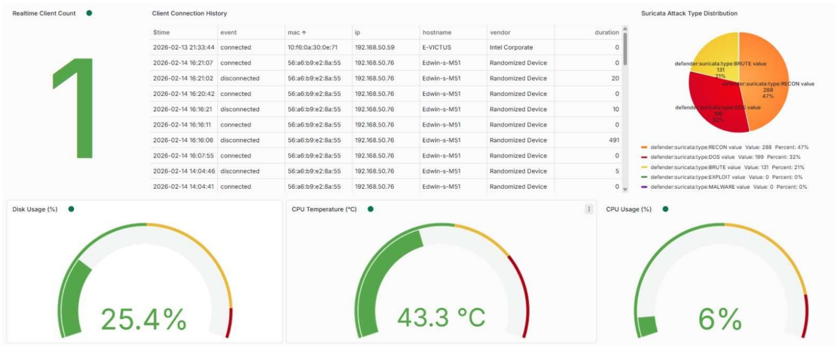


Figure 9. DefenderPi Security Operations Dashboard

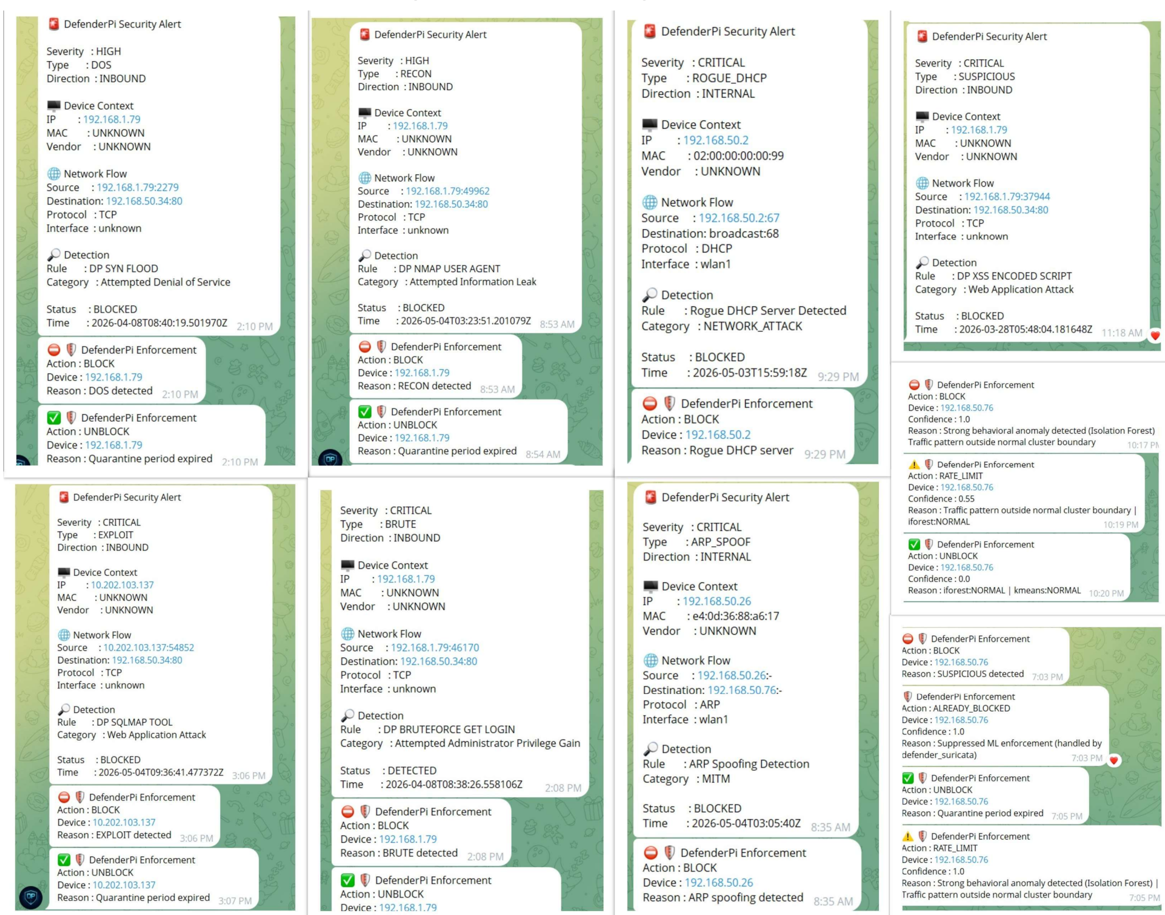
Real-Time Alerting

The platform integrates with Telegram to provide immediate notifications whenever suspicious activity is detected.

Alerts include:

- Source IP
- Threat Type
- Detection Time
- Severity Level
- Response Status

This enables remote monitoring without requiring direct access the dashboard.



to

Figure 11. Real-Time Security Alert Notification

Threat Intelligence Integration

To support investigation and incident response, DefenderPi integrates external intelligence services.

Supported Sources:

- VirusTotal, AbuseIPDB
- Google Safe Browsing
- URLhaus
- HaveIBeenPwned

Administrators can perform enrichment directly through Telegram commands.

Figure 12. Threat Intelligence and OSINT Integration

Engineering Challenges and Optimizations

1. Resource Optimization on Raspberry Pi

Challenge:
Integrating Suricata IPS, DNS filtering, behavioral analytics, dashboards, alerting, and policy enforcement on a Raspberry Pi initially caused high CPU and memory utilization, resulting in instability and occasional system hangs.
Solution:
System components were profiled and optimized by reducing unnecessary processing, minimizing logging overhead, improving event handling, and optimizing service interactions. These improvements enabled stable 24×7 operation with low resource consumption.

2. Suricata and Inline IPS Tuning

Challenge:
Running Suricata in inline IPS mode introduced performance overhead, packet processing delays, and increased false positives due to large community rule sets and inefficient traffic handling configurations.
Solution:
Suricata was extensively tuned through rule-set optimization, threshold adjustments, and NFQUEUE configuration refinement. Traffic processing paths and network parameters were optimized to improve inspection efficiency while maintaining reliable detection and prevention capabilities.

3. Behavioral Analytics and Policy Correlation

Challenge:
Initial machine learning models generated inconsistent results due to feature selection challenges, while direct enforcement based on individual alerts increased false-positive risk.
Solution:
Multiple iterations of feature engineering and model retraining were performed to improve anomaly detection accuracy. A fusion-based policy engine was then developed to correlate Suricata alerts, DNS events, network monitoring data, and behavioral analytics before initiating enforcement actions, significantly reducing false positives.

4. Firewall Enforcement Architecture

Challenge:
Coordinating automated blocking, session termination, and dynamic firewall updates required careful rule ordering and interaction between multiple security modules.
Solution:
The firewall architecture was redesigned using iptables, ipset, and conntrack integration. Proper chain ordering and enforcement logic ensured reliable threat containment without conflicts between system components.

5. Project Recovery and Reimplementation

Challenge:
A storage failure during development resulted in the loss of several recently implemented components, while the latest available backup was several weeks old.
Solution:
Core modules were reconstructed from documentation, previous backups, and testing artifacts. The recovery process led to a cleaner, more modular implementation and reinforced the importance of version control, backup strategies, and deployment documentation.

Performance Results

DefenderPi was evaluated under multiple simulated attack scenarios.
The platform successfully detected and contained:

- DoS Attacks
- Port Scanning & Network Reconnaissance
- Brute-Force Attempts
- ARP Spoofing
- Rogue DHCP Activity
- Sql Injection
- XSS

Resource Utilization

- CPU Usage: Approximately 12–18%
- Memory Usage: Below 650 MB
- Stable Long-Term Operation
- No Service Interruptions During Testing

Detection Performance

Attack Type	Avg Detection Latency (s)	Avg Containment Time (s)
DOS	0.59	0.63
RECON	0.68	0.75
BRUTE	1.32	1.4

Figure 13. Detection and Containment Performance Analysis
Containment actions were executed within approximately 1.4 seconds of detection.

Conclusion

DefenderPi is an AI-assisted inline network defense platform that integrates intrusion prevention, behavioral analytics, DNS security, reputation-based enforcement and automated threat response into a single Raspberry Pi deployment. The project demonstrates practical expertise in network security, threat detection, automation, and incident response while delivering enterprise-inspired protection for small and edge networks.